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Foundations of Microeconomic Analysis & Core Principles

Chapter 1 — Curriculum Module

Department of Economics

November 2, 2026

Contents

1 Economics and Scarcity: The Fundamental Problem

Economics is the social science that analyzes how individuals, firms, governments, and societies allocate scarce resources to satisfy unlimited human wants. At the core of all economic inquiry lies the immutable condition of **scarcity**: the physical and temporal constraint that available resources are insufficient to produce all goods and services desired by humanity.

As defined by Lionel Robbins in his seminal treatise *An Essay on the Nature and Significance of Economic Science* (1932):

> *"Economics is the science which studies human behaviour as a relationship between ends and scarce means which have alternative uses."*

To produce economic output, societies utilize four fundamental factors of production:

1. **Land (Natural Resources):** All raw, uncultivated gifts of nature, including minerals, water, arable land, and atmospheric space.
2. **Labor (Human Effort):** The physical and mental exertions of individuals applied to the transformation of inputs into valuable output.
3. **Capital (Manufactured Assets):** Durable, human-made physical assets used in production, such as machinery, factories, computing infrastructure, and transport networks.
4. **Entrepreneurship (Risk & Innovation):** The organizing capability that combines land, labor, and capital, introducing technological innovations and accepting financial risk.

2 Opportunity Cost and Decision-Making

Because choosing one alternative necessitates forfeiting another, every economic choice incurs an **opportunity cost**. Formally, the opportunity cost of an action is the value of the single highest-valued alternative foregone when a choice is made.

Opportunity cost consists of two distinct components:

$$\text{Total Economic Cost} = \text{Explicit Costs} + \text{Implicit Costs} \tag{1}$$

- **Explicit Costs:** Direct out-of-pocket monetary expenditures associated with an action (e.g., tuition fees, raw material expenses, equipment leases).
- **Implicit Costs:** The non-monetary opportunity value of resources already owned or time deployed, such as foregone wages or foregone interest income.

2.1 Economic Profit vs. Accounting Profit

Accounting profit considers only explicit monetary receipts and disbursements:

$$\text{Accounting Profit} = \text{Total Revenue} - \text{Explicit Costs} \tag{2}$$

In contrast, economic profit subtracts both explicit and implicit costs:

$$\text{Economic Profit} = \text{Total Revenue} - (\text{Explicit Costs} + \text{Implicit Costs}) \tag{3}$$

A firm earning zero economic profit is making a **normal profit**, earning exactly what its inputs could command in their best alternative deployment.

2.2 The Sunk Cost Fallacy

A critical analytical boundary in rational choice theory is the treatment of **sunk costs**. A sunk cost is a historical expenditure that has already been incurred and cannot be recovered by any current or future decision. Because sunk costs are invariant across future decision paths, rational decision-makers ignore them.

> **Core Principle 1.2: Opportunity Cost**
>
> The true cost of any decision is the net benefit of the next-best alternative option foregone. Rational decision-making evaluates actions solely on their forward-looking marginal benefits and costs, completely ignoring unrecoverable sunk costs.

3 Rationality and Incentives

Microeconomic analysis assumes that economic agents behave **rationally**. In neo-classical economic theory, rational agents possess well-defined, transitive preferences and choose actions intended to maximize their net utility (for consumers) or economic profit (for firms), subject to budget and resource constraints.

3.1 Incentive Alignment and Unintended Consequences

An **incentive** is a monetary or non-monetary reward or penalty that alters the expected marginal benefit or marginal cost of an action. As Gary Becker (1976) emphasized, human behavior responds systematically to shifts in incentives.

However, poorly designed policy intervention can induce moral hazard or perverse incentives, commonly referred to as the *Cobra Effect* or the *Law of Unintended Consequences*.

> **Empirical Case Study: Congestion Pricing and Demand Management**
>
> **Context:** Urban traffic congestion represents an unpriced negative external cost on urban networks. In 1975, Singapore introduced the Area Licensing Scheme (ALS), updated in 1998 to Electronic Road Pricing (ERP), imposing dynamic toll charges on vehicles entering central business districts during peak hours.
>
> **Economic Analysis:** By internalizing the marginal congestion cost, the monetary toll increased the explicit marginal cost of driving relative to public transit.
>
> **Empirical Findings:** Empirical evaluations (Phang & Toh, 2004) demonstrated an immediate 45% reduction in peak-hour traffic flow and a 20% increase in average traffic speeds. The dynamic pricing mechanism aligned private incentives with social optimum efficiency.

> **Core Principle 1.1: Scarcity and Choice**
>
> Scarcity is universal and persistent. Because resources are finite while wants are non-binding, choice is mandatory. Resolving allocation problems requires society to establish allocation mechanisms—whether through market prices, command directives, or traditional customs.

4 Marginal Analysis: Thinking at the Margin

Economic optimization rarely involves all-or-nothing binary choices; rather, it involves incremental decisions at the **margin**. Decision-makers evaluate small incremental additions to existing activities.

4.1 Mathematical Formalization of Marginal Optimization

Let $B(Q)$ represent the total benefit of consuming or producing Q units of an activity, and let $C(Q)$ represent the total cost function, where both functions are assumed to be continuously differentiable. The net economic surplus function $\Pi(Q)$ is:

$$\Pi(Q) = B(Q) - C(Q) \tag{4}$$

To maximize net surplus $\Pi(Q)$, the first-order necessary condition (FOC) requires taking the derivative with respect to Q and setting it to zero:

$$\frac{d\Pi(Q)}{dQ} = \frac{dB(Q)}{dQ} - \frac{dC(Q)}{dQ} = 0 \tag{5}$$

Defining Marginal Benefit as $MB(Q) \equiv \frac{dB(Q)}{dQ}$ and Marginal Cost as $MC(Q) \equiv \frac{dC(Q)}{dQ}$, the optimization condition yields:

$$MB(Q^*) = MC(Q^*) \tag{6}$$

The second-order condition (SOC) for a maximum requires:

$$\frac{d^2\Pi(Q)}{dQ^2} = \frac{d^2B(Q)}{dQ^2} - \frac{d^2C(Q)}{dQ^2} < 0 \implies \frac{d(MB)}{dQ} < \frac{d(MC)}{dQ} \tag{7}$$

This condition holds under standard assumptions of diminishing marginal utility ($\frac{d(MB)}{dQ} < 0$) and increasing marginal cost ($\frac{d(MC)}{dQ} > 0$).

5 Specialization, Comparative Advantage, and Trade

Trade allows individuals, regions, and nations to consume beyond their domestic production possibilities frontier (PPF). The analytical foundation of international trade theory was established by David Ricardo in *On the Principles of Political Economy and Taxation* (1817).

5.1 Absolute vs. Comparative Advantage

- **Absolute Advantage:** The ability of an agent or nation to produce a given good using fewer total inputs (or higher output per unit input) than another producer.
- **Comparative Advantage:** The ability of an agent or nation to produce a good at a lower **opportunity cost** than another producer.

5.2 Numerical Demonstration of Gains from Trade

Consider two economies, Nation A and Nation B, producing two goods: Microchips (M) and Wheat (W). Table **??** illustrates output per labor-hour.

Table 1: Output per Labor Hour and Opportunity Costs

Economy	**Microchips (M)**	**Wheat (W)**	**Opp. Cost of $1M$**	**Opp. Cost of $1W$**
Nation A	10 units	5 units	$0.5\,W$	$2.0\,M$
Nation B	6 units	4 units	$0.67\,W$	$1.5\,M$

Nation A has an absolute advantage in both goods. However, Nation A's opportunity cost of producing $1M$ is $0.5W$, which is lower than Nation B's ($0.67W$). Thus, Nation A has a **comparative advantage** in Microchips. Conversely, Nation B's opportunity cost of producing $1W$ is $1.5M$, which is lower than Nation A's ($2.0M$), giving Nation B a **comparative advantage** in Wheat.

By specializing according to comparative advantage and trading at terms of trade t satisfying $0.5\,W < t < 0.67\,W$ per Microchip, both economies achieve consumption possibilities strictly outside their individual PPFs.

6 Economic Systems and Market Efficiency

Societies organize resource allocation through distinct economic systems:

1. **Command Economies:** Resource allocation is centralized via state planning authorities.
2. **Market Economies:** Resource allocation is decentralized, guided by private ownership and self-interested interaction through price signals.
3. **Mixed Economies:** Market mechanisms operate alongside state regulation and public good provision.

Friedrich Hayek (1945) observed that market prices act as an information aggregation mechanism, transmitting dispersed knowledge about scarcity without requiring central coordination.

6.1 The Invisible Hand Mechanism

Adam Smith (1776) articulated that in competitive markets, individuals pursuing their private interests are led "by an invisible hand" to promote social welfare, maximizing total economic surplus.

6.2 Marginal Analysis in Market Equilibrium

In a competitive market without externalities, the market demand curve reflects the aggregate Marginal Private Benefit (MB), while the supply curve reflects Marginal Private Cost (MC).

Figure **??** illustrates the marginal analysis decision rule $MB(Q^*) = MC(Q^*)$ in market equilibrium.

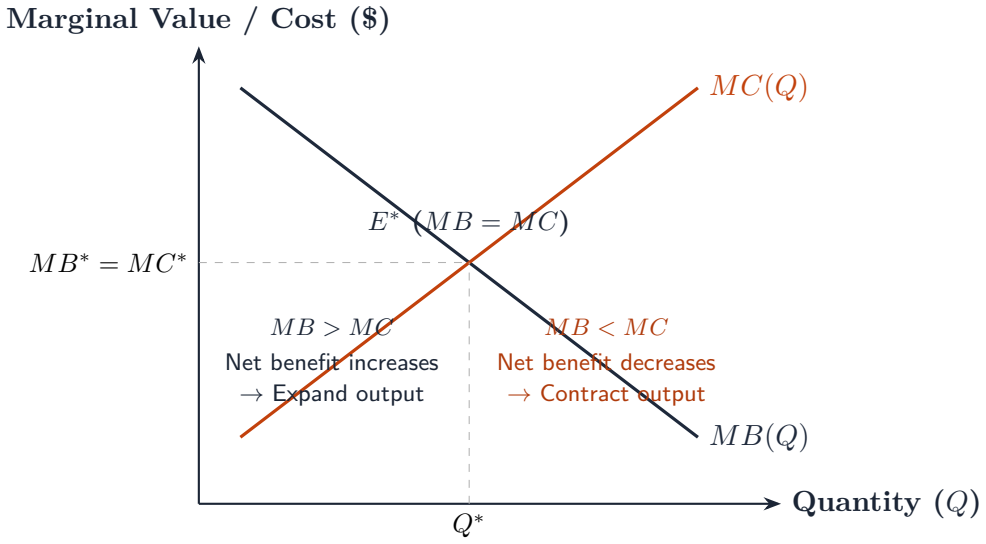


Figure 1: Marginal Analysis Optimization Rule ($MB = MC$)

At quantity $Q < Q^*$, $MB > MC$, meaning an additional unit adds more benefit than cost, increasing net social welfare. At $Q > Q^*$, $MC > MB$, implying overproduction where marginal cost exceeds marginal benefit. Equilibrium Q^* maximizes total surplus.

7 Microeconomics vs. Macroeconomics: Scope and Interconnection

Economic analysis is divided into two major branches:

- **Microeconomics:** The study of individual decision-making units—consumers, workers, firms, and specific market structures—and their interactions in individual markets.
- **Macroeconomics:** The study of aggregate economic phenomena, including economy-wide output (GDP), aggregate price inflation, unemployment rates, national debt, and monetary policies.

7.1 Microfoundations of Macroeconomics

Modern macroeconomic modeling relies explicitly on microfoundations. Following Robert Lucas's critique (1976), aggregate economic dynamics are derived by aggregating the utility-maximizing choices of forward-looking microeconomic agents under explicit policy regimes.

8 Positive vs. Normative Analysis

Economists distinguish strictly between descriptive scientific analysis and value-based policy evaluation:

- **Positive Economics:** Objective, descriptive statements about how the economy functions. Positive statements are empirical claims capable of being verified or falsified by data (e.g., *"A 10% increase in minimum wage reduces youth employment by 1.5%"*).
- **Normative Economics:** Prescriptive statements reflecting subjective values, ethical judgments, or political priorities regarding how the economy *ought* to be structured (e.g., *"The minimum wage should be raised to ensure a living wage"*).

Methodological rigor requires economists to state positive empirical mechanisms clearly, leaving normative assessments to public policy decision-makers.

9 Economic Models and Empirical Foundations

To isolate causal mechanisms in complex socio-economic systems, economists construct **economic models**. Models are simplified abstractions of reality based on explicit assumptions.

9.1 The *Ceteris Paribus* Assumption

The methodological anchor of economic modeling is the ***ceteris paribus*** assumption ("all other things held constant"). By holding confounding variables fixed, economists isolate the specific marginal effect of changing a single explanatory variable.

9.2 Empirical Identification and Econometrics

As emphasized by Joshua Angrist and Jörn-Steffen Pischke (2009), modern economics relies on causal inference techniques to test positive models against empirical data:

1. **Randomized Controlled Trials (RCTs):** Field experiments assigning treatment and control groups to evaluate policy efficacy (Banerjee, Duflo, & Kremer, 2019).
2. **Difference-in-Differences (DiD):** Exploiting spatial or temporal policy variations across treatment and comparison regions.
3. **Instrumental Variables (IV):** Utilizing exogenous variation to address endogeneity and omitted variable bias.

Key Takeaways

1. **Scarcity & Choice:** Resources are scarce; unlimited wants necessitate choice and allocation mechanisms.
2. **Opportunity Cost:** Every choice incurs an opportunity cost representing the next-best foregone alternative.
3. **Marginal Decision Rule:** Optimization occurs where Marginal Benefit equals Marginal Cost ($MB = MC$).
4. **Comparative Advantage:** Specialization and trade according to relative opportunity costs produce mutual gains.
5. **Positive vs. Normative:** Positive analysis evaluates testable causal facts; normative analysis reflects value judgments.